Information Systems III Chapter 10
Concurrency Control and Database Recovery
Section A: Multiple Choice

1. A transaction is in _____ state after the final statement has been executed
   a) Partially committed
   b) Active
   c) Committed
   d) None of the above
2. _____ ensures that once transaction changes are done, they cannot be undone or lost, even in the event of a system failure.
   a) Atomicity
   b) Consistency
   c) Durability
   d) Isolation
3. A database management system (DBMS) uses a transaction _____ to keep track of all transactions that update the database.
   a) Log
   b) Table
   c) Block
   d) Statement
4. Although the _____ locking approach improves the availability of data. Its management requires high overhead.
   a) Database level
   b) Table level
   c) Page level
   d) Row level
5. A _____ database state is one in which all data integrity constraints are satisfied.
   a) Fixed
   b) Complete
   c) Consistent
   d) Secure
6. If a transaction T1 is being executed and is using the data item X, and that data item cannot be accessed by any other transaction (T2…. Tn) until T1 ends. This is an example of which transaction property?
   a) Atomicity
   b) Consistency
   c) Isolation
   d) Durability
7. A transaction normally terminates when it reaches which of the following?
   a) COMMIT
   b) ROLLBACK/ABORT
   c) A failure that causes it to abort

d) All of the above

8. A _____ lock is good for batch processes, but it is unsuitable for multiuser DBMS.
   a) Field level
   b) Row level
   c) Database level
   d) Table level
9. All lock information is managed by a _____ , which is responsible for assigning and policing the locks used by the transactions.
   a) Scheduler
   b) DBMS
   c) Lock manager
   d) Locking agent
10. Transaction support is provided by two SQL statements, namely:
   a) DONE and ROLLBACK
   b) DONE and QUIT
   c) COMMIT and ROLLBACK
   d) COMMIT and END
11. The _____ also make sure that the computer's central processing unit (CPU) and storage systems are used efficiently.
   a) Time stamping
   b) Checkpoint
   c) Scheduler
   d) Lock
12. How does the scheduler ensure serializability?
   a) By executing transactions randomly
   b) By enforcing a strict order of operations
   c) By ignoring transaction conflict
   d) By disabling concurrent access

Section B: True/False

1. A table-level lock is good for batch processes, but it is unsuitable for multiuser DBMSs:
2. A checkpoint is an operation in which the DBMS writes updated (dirty) buffers from memory to disk.
3. In a deadlock avoidance technique, a transaction can begin execution without acquiring all the necessary locks.
4. Monotonicity ensures that timestamp values always increase.
5. The transaction log keeps track of all transactions that modify the database.
6. When the recovery uses the deferred-write technique, the transaction operations do not immediately update the physical database.
7. The write-ahead-log protocol ensures that transaction logs are always written after the database data has been updated.
8. A database request is the equivalent of a single SQL statement in an application program or transaction

9. Serializability occurs when concurrent execution of several transactions yields consistent results.
10. All transactions update the database.

Section C: Short Questions

1. Describe row-level locking in a database system and mention one advantage of using it. (3)
2. Describe how the stamping methods are used for concurrency control. (3)
3. Let's say you are a database administrator in a multi-user or shared database. Briefly explain the problem of inconsistent retrievals that you may encounter if you fail to maintain concurrency control and give an example to illustrate this phenomenon. (4)
4. T1 has timestamp 10 and T2 has timestamp 20. T2 is holding a lock required by T1. Using the Wait/Die scheme, state what happens and explain why. (4)
5. T1 has timestamp 10, and T2 has timestamp 20. T2 is holding a lock required by T1. Using the Wound/Wait scheme, state what happens and explain why. (4)
6. Differentiate between deferred-write and write-through recovery techniques. (6)
7. Consider the following transaction:

T1 START
UPDATE Account A
UPDATE Account B
SYSTEM FAILURE
T1 had not reached COMMIT when the failure occurred.

a) What happens to T1? (1)
b) What recovery operation is required? (1)
c) Explain why this operation is necessary. (2)
d) What recovery information can the DBMS use to determine what happened? (1)